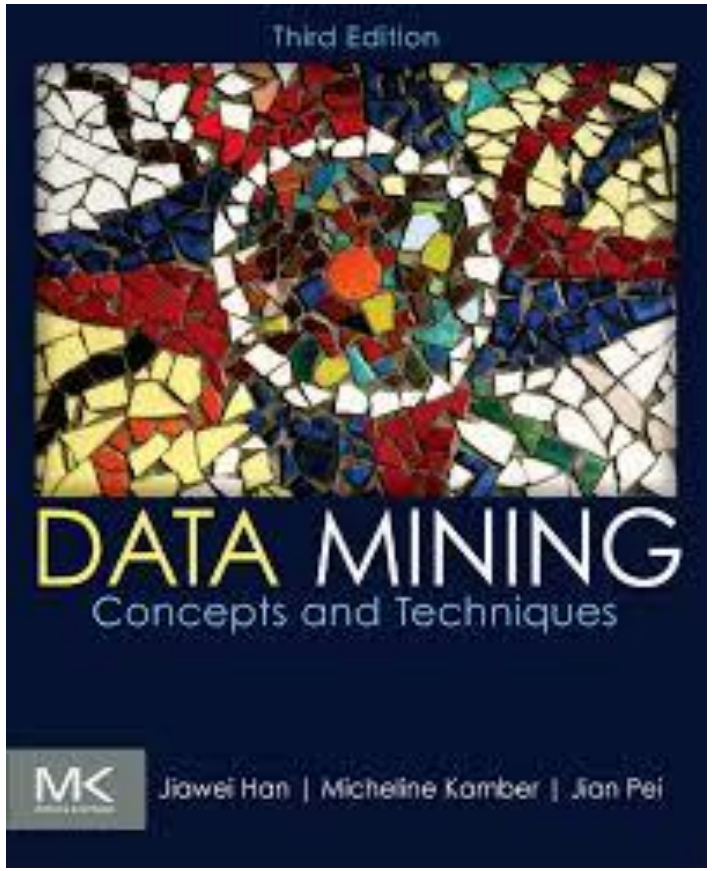
Machine Learning and Data Mining Basics

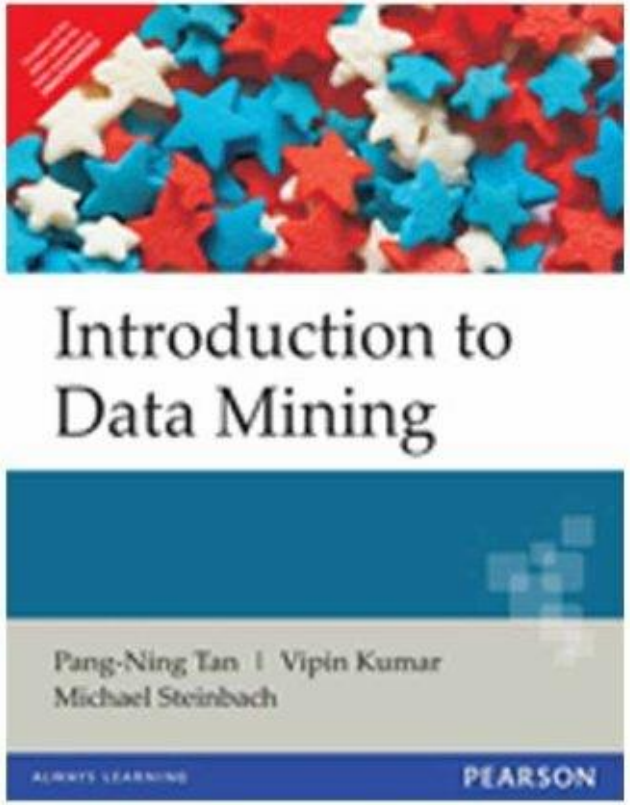
Course Goal

- **Differentiate Data Mining and Machine Learning** – Understand the scope, purpose, and key differences.
- **Understand Data Representation** – Learn how data is structured, formatted, and encoded for ML.
- **Master Data Preprocessing** – Handle missing data, scaling, normalization, and feature engineering.
- **Comprehend the Machine Learning Life Cycle** – From problem definition to model deployment.
- **Identify Types of Machine Learning** – Supervised, Semi-supervised, Unsupervised, and Reinforcement Learning.
- **Implement Core ML Algorithms** – Linear & logistic regression, decision trees, SVM, KNN, Naïve Bayes, etc.
- **Evaluate ML Models** – Metrics, validation, cross-validation, and performance improvement techniques.
- **Introduction to Deep Learning** – Basics of neural networks and their applications.
- **Develop Practical Thinking Skills** – Learn frameworks for problem-solving and common beginner pitfalls.

Course Materials and References

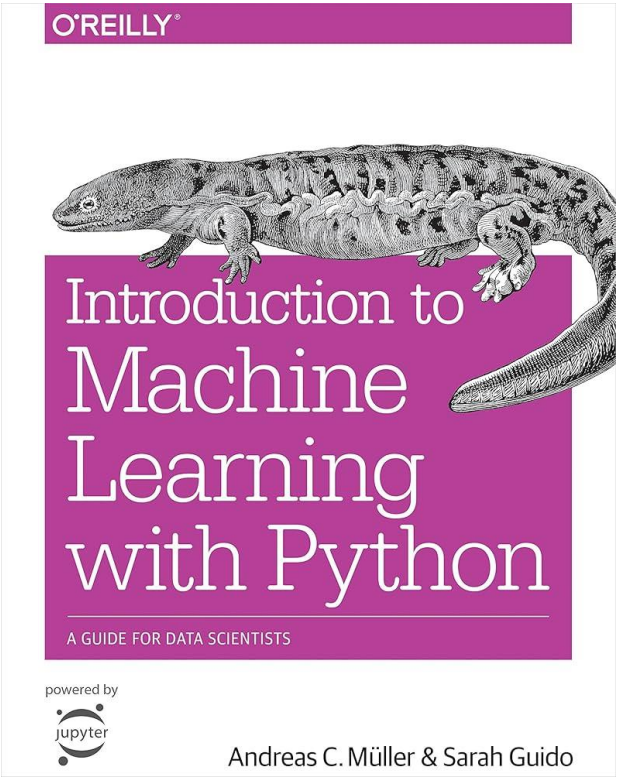


Data Mining Concepts and Techniques by Jiawei Han, Micheline Kamber, Jian Pei ·

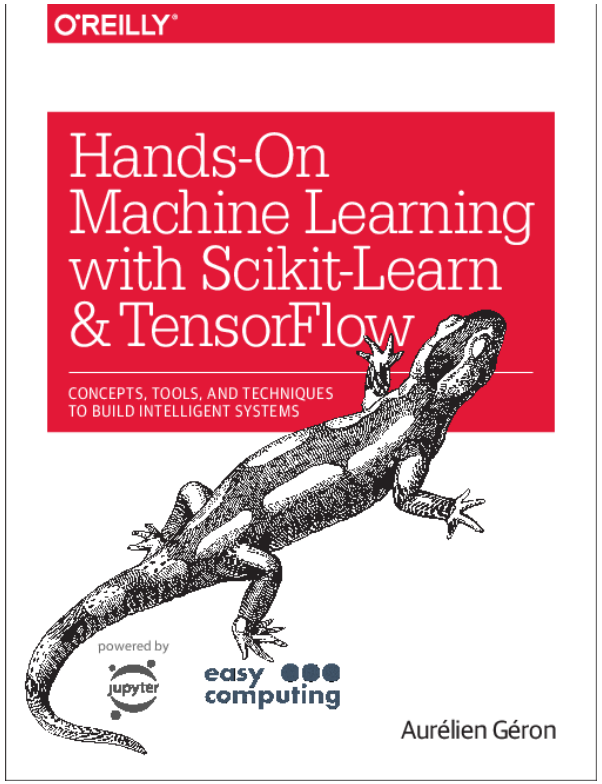


Introduction to Data Mining by Pang-Ning Tan Michael Steinbach Vipin Kumar

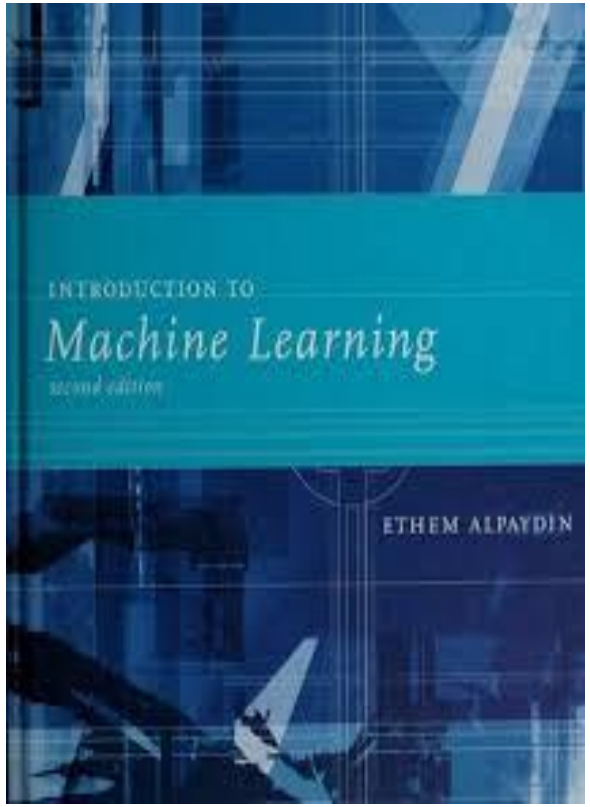
Course Materials and References



Introduction to Machine Learning with Python by Andreas C. Muller and Sarah Guido



Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurélien Géron



Introduction to Machine Learning by Ethem Alpaydın

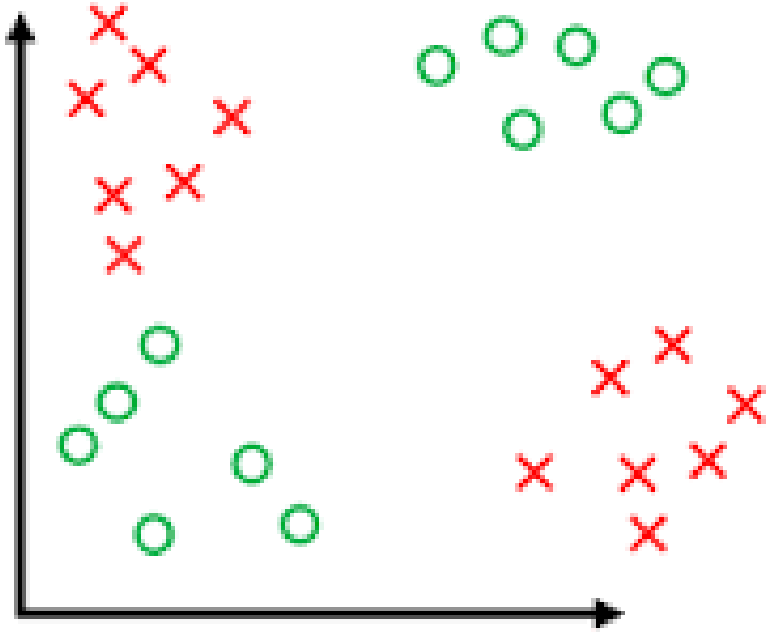
Outline

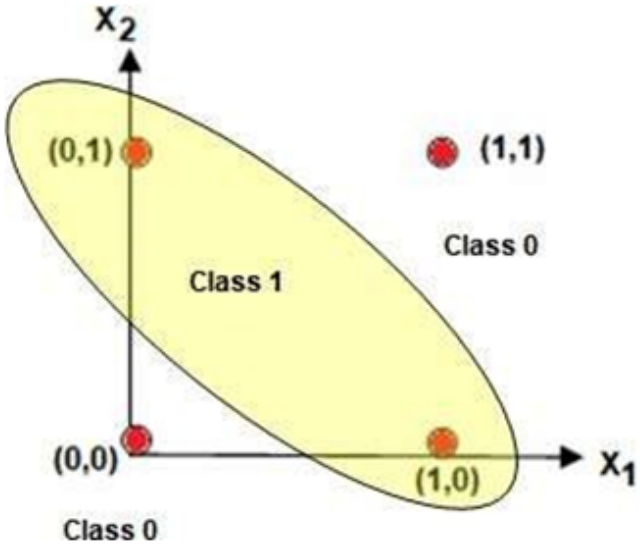
- NOT linearly separable Problems
- Decision Trees
- Decision Tree Terminology
- Impurity
- Tree Pruning

- Naïve Bayes
- Bayes
- Bayes' Theorem
- Numerical Example
- Advantages of Naive Bayes
- Disadvantages of Naive Bayes

What if the data is NOT linearly separable?

Logistic regression gives one straight boundary.





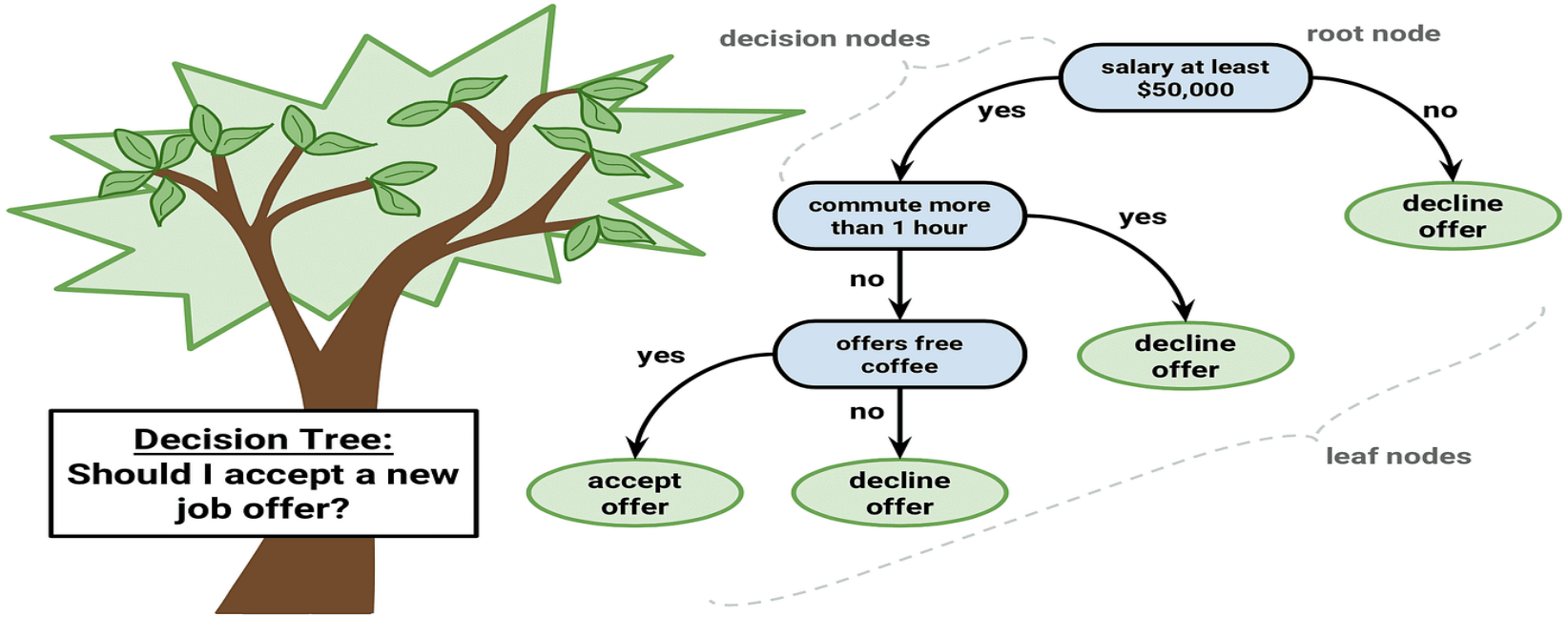
Truth Table for XOR Gate		
X1	X2	Output Class
0	0	0
0	1	1
1	0	1
1	1	0

What if we want boundaries that bend?

Decision Trees

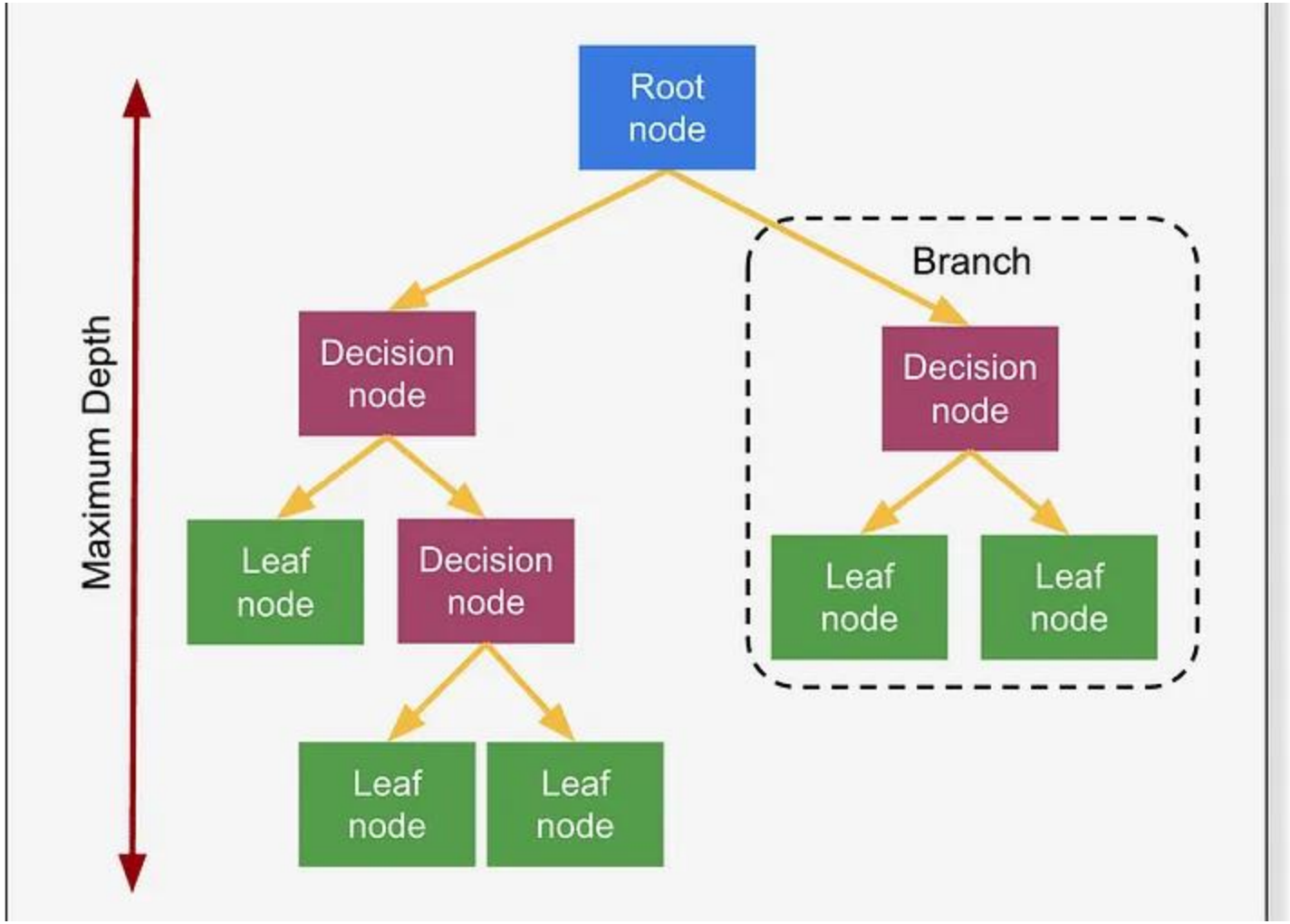
A decision tree is a graphical representation of all the possible solutions to a decision based on a certain conditions.

A sequence of questions.



It’s just nested if-else rules.

Decision Tree Terminology



Decision Tree Terminology

- **Root Node:**
  - Represents the entire dataset
- **Decision Nodes:**
  - The nodes where the dataset is further split based on a certain feature.
- **Leaf Nodes:**
  - They represent the final output or the decision of the model.
  - They do not split further and contain the outcome of the decision path.
- **Branch:**
  - The connection between nodes, representing the outcome of a test.
- **Parent Node**
  - Any node that splits into one or more branches
- **Child Node**
  - - Created from a parent node after a split

Algorithm for Decision Tree Induction

- Basic algorithm (a greedy algorithm)
  - Tree is constructed in a top-down recursive divide-and-conquer manner
  - At start, all the training examples are at the root
  - Attributes are categorical (if continuous-valued, they are discretized in advance)
  - Examples are partitioned recursively based on selected attributes
  - Test attributes are selected on the basis of a heuristic or statistical measure (e.g., information gain)
- Conditions for stopping partitioning
  - All samples for a given node belong to the same class
  - There are no remaining attributes for further partitioning – majority voting is employed for classifying the leaf
  - There are no samples left

Example

Goal: Predict whether someone will like go to **:**.

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W2	Sunny	No	Rich	Tennis
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W5	Rainy	No	Rich	Stay In
W6	Rainy	Yes	Poor	Cinema
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping
W9	Windy	Yes	Rich	Cinema
W10	Sunny	No	Rich	Tennis

There are 4 possible outputs:

Cinema, Tennis,
Stay In, Shopping.

6 instances of Cinema
2 instances of Tennis
1 instance of Stay In
1 instance of Shopping

What Is Impurity?

Compute the **Gini Index** for the overall collection of training samples:

- A node is **pure** if everyone in it has the same label.
- A node is **impure** if it mixes Yes and No

❖ **Gini impurity:**

$$G = 1 - \sum_{i=1}^{n} (p_i)^2$$

$$Gini = 1 - [(\frac{6}{10})^2 + (\frac{2}{10})^2 + (\frac{1}{10})^2 + (\frac{1}{10})^2] = 0.58$$

- 0 means perfect purity
- 0.5 means maximum impurity (for binary classification).
- The higher the Entropy (or Gini Index), the more random (mixed) the data is

Compute Root Impurity

Computation of Gini index for money attribute:

Has two possible values: Rich (7 Instances) , Poor (3 Instances)

For Money=Poor ⟹ 3 “Cinema”

$$G(s) = 1 - [(\frac{3}{3})^2] = 0$$

For Money=Rich ⟹ 3 “Cinema” , 2 “Tennis”, 1 “Stay In”,1 “Shopping”

$$G(s) = 1 - [(\frac{3}{7})^2 + (\frac{2}{7})^2 + (\frac{1}{7})^2 + (\frac{1}{7})^2] = 0.694$$

Weighted Average Gini ⟹ $= \left(\frac{N_{left}}{N_{total}}\right) * G_{left} + \left(\frac{N_{right}}{N_{total}}\right) * G_{right}$

Weighted Average Gini (Money) $= \frac{3}{10} * 0 + \frac{7}{10} * 0.694 = 0.486$

- Same For Parents and Weather

Weather Has Three possible values: Sunny (3 Instances) , Windy (4 Instances), Rainy (3 Instances)

For Weather = Sunny ⇒ 1 “Cinema” and 2 “Tennis”

$$G(s) = 1 - [(\frac{2}{3})^2 + (\frac{1}{3})^2] = 0.444$$

For Weather=Windy ⇒ 3 “Cinema” , 1 “Shopping”

$$G(s) = 1 - [(\frac{3}{4})^2 + (\frac{1}{4})^2] = 0.375$$

For Weather=Rainy ⇒ 2 “Cinema” , 1 “Stay In”

$$G(s) = 1 - [(\frac{2}{3})^2 + (\frac{1}{3})^2] = 0.444$$

- Weighted Average Gini (Weather) $= \frac{3}{10} * 0.444 + \frac{4}{10} * 0.375 + \frac{3}{10} * 0.444 = 0.416$

- For Money - Gini index $= \frac{3}{10} * 0 + \frac{7}{10} * 0.694 = 0.486$
- For Weather - Gini index $= \frac{3}{10} * 0.444 + \frac{4}{10} * 0.375 + \frac{3}{10} * 0.444 = 0.416$
- For Parents - Gini index $= \frac{5}{10} * 0 + \frac{5}{10} * 0.72 = 0.36$ ✓

Parents is selected as it has the smallest Gini Index

Parents

Yes

Weekend	Weather	Parents	Money	Decision
W1	Sunny	Yes	Rich	Cinema
W3	Windy	Yes	Rich	Cinema
W4	Rainy	Yes	Poor	Cinema
W6	Rainy	Yes	Poor	Cinema
W9	Windy	Yes	Rich	Cinema

No

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W5	Windy	No	Rich	Stay In
W7	Rainy	No	Poor	Cinema
W8	Rainy	No	Rich	Shopping
W10	Windy	No	Rich	Tennis

No computation is needed here since it is always Cinema.

Computation for Gini Index for Parents=No | Weather Attribute

- Weather =Sunny, 2 Instances both are "Tennis"

$$G(s) = 1 - [(\frac{2}{2})^2] = 0$$

- Weather =Rainy, 1 "Stay In"

$$G(s) = 1 - [(\frac{1}{1})^2] = 0$$

- Weather =Windy, 1 "Cinema" , 1 "Shopping"

$$G(s) = 1 - [(\frac{1}{2})^2 + (\frac{1}{2})^2] = 0.5$$

- Weighted Average Gini (Parents=0|Weather) $= \frac{2}{5} * 0 + \frac{1}{5} * 0 + \frac{2}{5} * 0.5 = 0.2$

Computation for Gini Index for Parents=No | Money Attribute

- Money = Rich, 2 "Tennis" , 1 "Stay In", 1 "Shopping"

$$G(s) = 1 - [(\frac{2}{4})^2 + (\frac{1}{4})^2 + (\frac{1}{4})^2 +] = 0.625$$

- Money = Poor, 1 "Cinema"

$$G(s) = 1 - [(\frac{1}{1})^2] = 0$$

- Weighted Average Gini (Parents=0|Money) $= \frac{4}{5} * 0.625 + \frac{1}{5} * 0 = 0.5$

- For Parents= No|Weather – Gini index = 0.2 ✓
- For Parents= No|Money – Gini index = 0.5

Weather is selected as it has the smallest Gini index

Weekend	Weather	Parents	Money	Decision
W2	Sunny	No	Rich	Tennis
W10	Sunny	No	Rich	Tennis

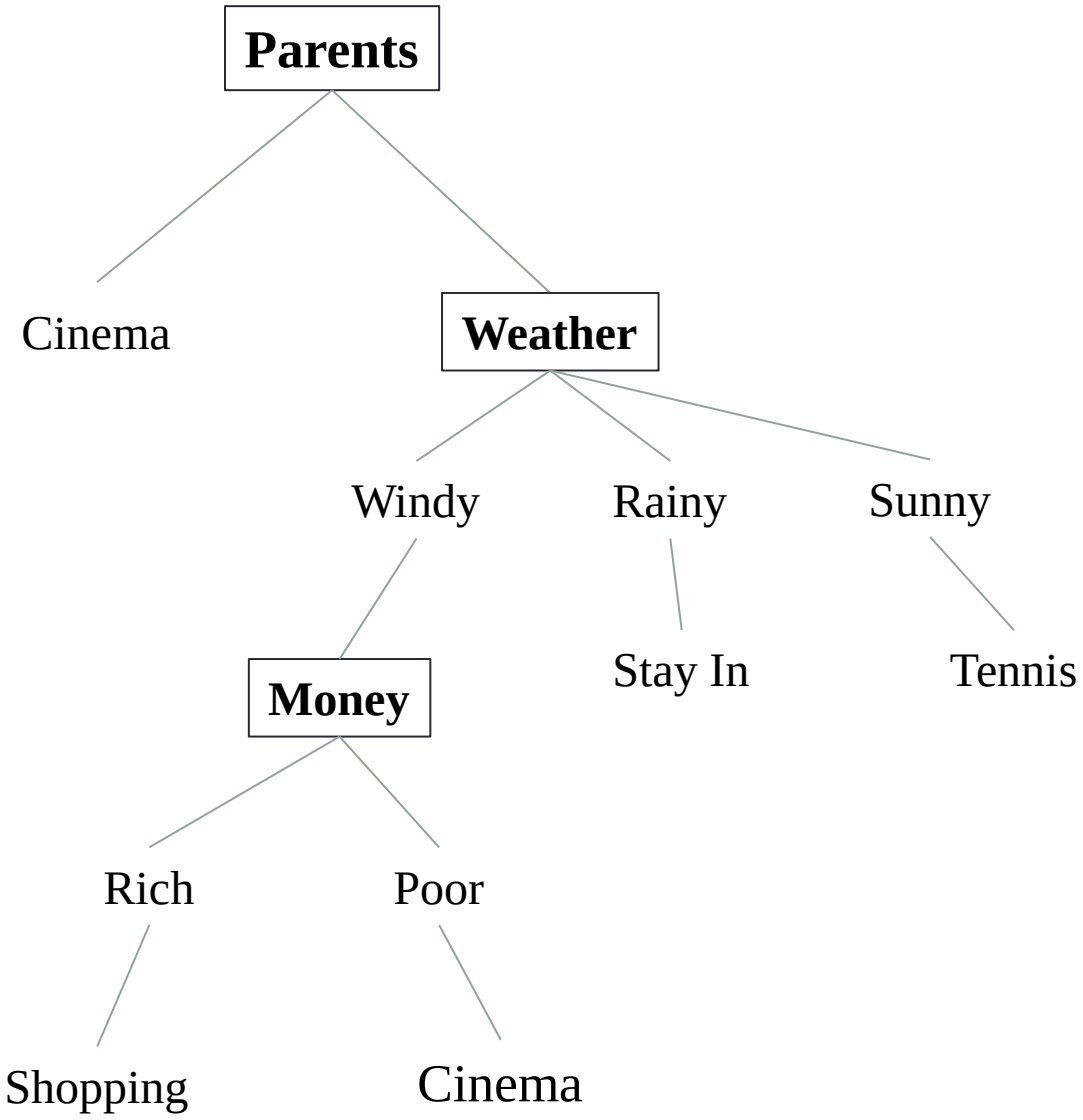
For Parent =No & Weather =Sunny , All instances are Tennis

For Parent =No & Weather =Rainy , All instances are Stay In

Weekend	Weather	Parents	Money	Decision
W5	Rainy	No	Rich	Stay In

Weekend	Weather	Parents	Money	Decision
W7	Windy	No	Poor	Cinema
W8	Windy	No	Rich	Shopping

For Parent =No & Weather =Windy, We need to Split.


Parents
Cinema
Weather
Windy
Rainy
Sunny
Money
Stay In
Tennis
Rich
Poor
Shopping
Cinema

- Other statistical measure is information gain using Entropy

$$\text{Entropy} = -\ \Sigma\ (p_i\ * log_2(p_i))$$

For binary classification, the upper bound of Entropy is 1.

- A tree can grow until it perfectly fits the training data, which often hurts performance on new data.

This introduces:

- Overfitting : Very deep tree → memorizes training data
- Underfitting : Very shallow tree → misses patterns

Tree Pruning

- To fix overfitting we use Pruning

- Pre-pruning (early stopping)

Stop growing the tree when:

  - Depth > threshold
  - Samples < threshold

- Post-pruning;

Grow full tree → then remove weak branches.

Naïve Bayes

- A probabilistic classifier
- Predicts the category of a data point using Bayes' Theorem
- Assumes that all features are independent of each other (Naïve Assumption) → Each feature contributes to the predictions with no relation between each other.
- Used in text classification and sentiment analysis.

Principle:

- Calculates the probability of a data point belonging to each class based on the observed features.
- Then predicts the class with the highest probability as the final prediction.

Assumption of Naive Bayes

- **Feature independence:**

  We assume that each feature (or piece of information) in the data does not affect any other feature. In reality, this assumption isn't always true.

- **Continuous features are normally distributed:**

  If a feature is continuous, then it is assumed to be normally distributed within each class.

- **Discrete features have multinomial distributions:**

  If a feature is discrete, then it is assumed to have a multinomial distribution within each class.

- **Features are equally important:**

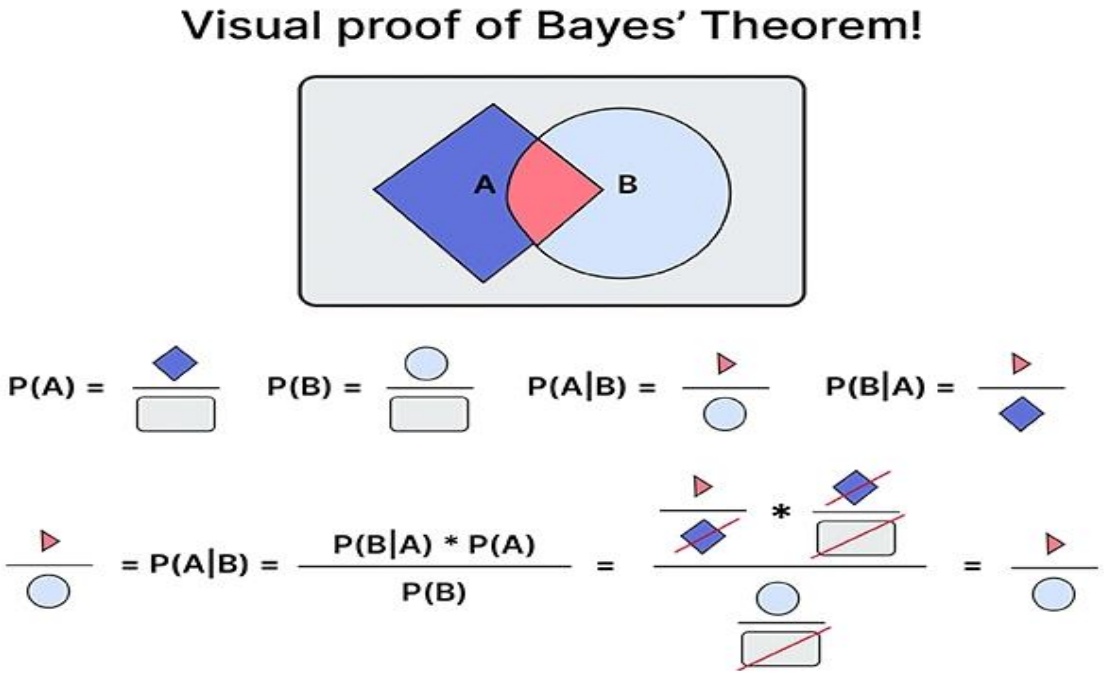
  All features are assumed to contribute equally to the prediction of the class label.

- **No missing data:**

  The data should not contain any missing values.

Bayes' Theorem

- Helps in calculating conditional probabilities.
- Conditional probability: is the probability of an event occurring given that another event has already occurred.



Definition

It is defined as:

$$P(X \mid Y) = \frac{P(Y \mid X) \cdot P(X)}{P(Y)}$$

Where:

- $P(X/Y)$: conditional probability of event $\boldsymbol{X}$ occurring given that event $\boldsymbol{Y}$ has occurred (posterior probability)
- $P(Y/X)$: probability of event $\boldsymbol{Y}$ occurring given that event A has already occurred (likelihood)
- $P(X)$: prior probability of event $\boldsymbol{X}$, before considering any new evidence.
- $P(Y)$: probability of event $\boldsymbol{Y}$.

Using the chain rule, the likelihood P(X | y) can be decomposed as:

$$P(X|y) = P(x_1, x_2, \ldots, x_n|y)$$

$$= P(x_1|x_2, \ldots, x_n, y) * P(x_2|x_3, \ldots, x_n, y) \ldots P(x_n|y)$$

But because of the Naive's conditional independence assumption, the conditional probabilities are independent of each other

$$P(X|y) = P(x_1|y) * P(x_2|y) \ldots P(x_n|y)$$

- Thus, by conditional independence, we have:

$$P(y|X) = \frac{P(x_1|y)*P(x_2|y)\ldots\ldots P(x_n|y)*P(y)}{P(x_1)*P(x_2)\ldots.P(x_n)}$$

And as denominator remains constant for all values, the posterior probability can then be:

$$P(y|x_1, x_2, \ldots., x_n) \propto P(y)\prod_{i=1}^{n} P(x_i|y)$$

Example

Given Outlook, Temperature, Humidity and Wind Information, we want to carry out prediction for Play: Yes or No.

- Mathematically, which one is greater
  - P(Play Yes | Outlook, Temp., Humidity, Wind)
  - P(Play No | Outlook, Temp., Humidity, Wind)
- Predict for Sunny outlook, High humidity, Cool temperature and Weak wind.
- Predict the most likely.

Day	Outlook	Temp.	Humidity	Wind	Play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Calculate the frequency table for all features

Day	Outlook	Temp.	Humidity	Wind	Play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Training

Frequency table		**Play**	
		Yes	No
Outlook	Sunny		
	Overcast		
	Rain		

Frequency table		**Play**	
		Yes	No
Temp	Hot		
	Mild		
	Cold		

Frequency table		**Play**	
		Yes	No
Humidity	High		
	Normal		

Frequency table		**Play**	
		Yes	No
Wind	Storng		
	Weak		

Calculate the frequency table for all features

Day	Outlook	Temp.	Humidity	Wind	Play
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Training

Frequency Table		Play	
		Yes	No
Outlook	Sunny	2	3
	Overcast	4	0
	Rain	3	2

Frequency Table		Play	
		Yes	No
Temp	Hot	2	2
	Mild	4	2
	Cold	3	1

Frequency Table		Play	
		Yes	No
Humidity	High	3	4
	Normal	6	1

Frequency Table		Play	
		Yes	No
Wind	Strong	3	3
	Weak	6	2

Calculate the likelihood table for all features

Likelihood table		Play		
		Yes	No	
Outlook	Sunny	2/9	3/5	5/14
	Overcast	4/9	0/5	4/14
	Rain	3/9	2/5	5/14
		9/14	5/14	

$P(X|c) = P(Sunny \mid Yes) = 2/9 = 0.22$

$P(X) = P(Sunny) = 5/14 = 0.35$

$P(c) = P(Yes) = 9/14 = 0.64$

Likelihood of "Yes" given sunny is:

$P(C|X) = P(Yes|Sunny) = P(Sunny|Yes) * P(Yes) / P(Sunny) = (0.22*0.64)/0.35 = 0.40$

Likelihood of "No" given sunny is:

$P(C|X) = P(No|Sunny) = P(Sunny|No) * P(No) / P(Sunny) = (0.6*0.35)/0.35 = 0.60$

Frequency Table		Play		
		Yes	No	
Temp	Hot	2/9	2/5	4/14
	Mild	4/9	2/5	6/14
	Cold	3/9	1/5	4/14
		9/14	5/14	

Frequency Table		Play		
		Yes	No	
Humidity	High	3/9	4/5	7/14
	Normal	6/9	1/5	7/14
		9/14	5/14	

P(Yes|Cold) = P(Cold|Yes) * P(Yes) / P(Cold)
= (0.33*0.64)/0.28 = 0.75
P(No|Cold) = (0.2*0.35)/0.28 = 0.25

P(Yes|High) = (0.33*0.64)/0.5 = 0.42
P(No|High) = (0.8*0.35)/0.5 = 0.56

Frequency Table		Play		
		Yes	No	
Wind	Strong	3/9	3/5	6/14
	Weak	6/9	2/5	8/14
		9/14	5/14	

P(Yes|Weak) = (0.66*0.64)/0.57= 0.74
P(No|Weak) = (0.4*0.35)/0.57 = 0.25

Test Data:

$$P(\text{Play} = \text{Yes} \mid \text{Outlook} = \text{Sunny},\ \text{Temp} = \text{Cool},\ \text{Humidity} = \text{High},\ \text{Wind} = \text{Weak})$$

$$= \frac{P(\text{Outlook} = \text{Sunny},\ \text{Temp} = \text{Cool},\ \text{Humidity} = \text{High},\ \text{Wind} = \text{Strong} \mid \text{Play} = \text{Yes})\, P(\text{Play} = \text{Yes})}{P(\text{Outlook} = \text{Sunny},\ \text{Temp} = \text{Cool},\ \text{Humidity} = \text{High},\ \text{Wind} = \text{Strong})}$$

Naïve Assumption:

- Feature are mutually independent given the label!

P(Outlook Sunny, Temp Cool, Humidity High, Wind = Strong | Play = Yes)

= P(Outlook=Sunny|Play = Yes) P(Temp= Cool|Play = Yes) P(Humidity= High|Play = Yes) P(Wind= Strong|Play = Yes)

$$P(\text{Outlook} = \text{Sunny} \mid \text{Play} = \text{Yes})\, P(\text{Temp} = \text{Cool} \mid \text{Play} = \text{Yes})\, P(\text{Humidity} = \text{High} \mid \text{Play} = \text{Yes})\, P(\text{Wind} = \text{Strong} \mid \text{Play} = \text{Yes})$$

$$\times P(\text{Play} = \text{Yes}) \qquad = \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} \qquad = 0.0053$$

$$P(\text{Outlook} = \text{Sunny} \mid \text{Play} = \text{No})\, P(\text{Temp} = \text{Cool} \mid \text{Play} = \text{No})\, P(\text{Humidity} = \text{High} \mid \text{Play} = \text{No})\, P(\text{Wind} = \text{Strong} \mid \text{Play} = \text{No})$$

$$\times P(\text{Play} = \text{No}) \qquad = \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} \qquad = 0.0206$$

$$P(\text{Play} = \text{Yes} \mid \text{Outlook} = \text{Sunny},\ \text{Temp} = \text{Cool},\ \text{Humidity} = \text{High},\ \text{Wind} = \text{Strong}) = \frac{0.0053}{0.0053 + 0.0206} = 0.2046$$

Normalization

$$P(\text{Play} = \text{No} \mid \text{Outlook} = \text{Sunny},\ \text{Temp} = \text{Cool},\ \text{Humidity} = \text{High},\ \text{Wind} = \text{Strong}) = \frac{0.0206}{0.0053 + 0.0206} = 0.7954$$

Play = No is more likely!

Advantages of Naive Bayes

1. This algorithm works very fast and can easily predict the class of a test dataset.
2. You can use it to solve multi-class prediction problems as it’s quite useful with them.
3. Naive Bayes classifier performs better than other models with less training data if the assumption of independence of features holds.
4. If you have categorical input variables, the Naive Bayes algorithm performs exceptionally well in comparison to numerical variables.

Disadvantages of Naive Bayes

1. If your test data set has a categorical variable of a category that wasn't present in the training data set, the Naïve Bayes model will assign it zero probability and won't be able to make any predictions in this regard.

This phenomenon is called 'Zero Frequency,' and you'll have to use a smoothing technique to solve this problem.

2. It assumes that all the features are independent. While it might sound great in theory, in real life, you'll hardly find a set of independent features.